

SIMULATION EXPERIMENT-01

SIMULATION-BASED ELECTROMAGNETIC ANALYSIS OF TEM, TE AND TM PROPAGATION MODES IN MICROWAVE TRANSMISSION STRUCTURES

Objectives of this Experiment:

1. To develop electromagnetic simulation models for TEM, TE and TM propagation modes in suitable microwave transmission structures.
2. To visualize and analyze the electric-field and magnetic-field distributions associated with different propagation modes.
3. To determine the cutoff frequency, propagation constant and field characteristics of selected TE and TM modes.
4. To compare the propagation behavior of different modes and identify their suitability for practical microwave transmission.

PROGRAM

```
clc;
clear;
close all;
a = 22.86e-3;
b = 10.16e-3;
c = 3e8;
f = 10e9;
lambda = c/f;
k0 = 2*pi/lambda;
fprintf('-----\n');
fprintf(' TEM, TE and TM MODE ANALYSIS\n');
fprintf('-----\n');
fprintf('Operating Frequency = %.2f GHz\n',f/1e9);
fprintf('Waveguide Width    = %.2f mm\n',a*1e3);
fprintf('Waveguide Height   = %.2f mm\n',b*1e3);
fprintf('Wavelength        = %.2f mm\n',lambda*1e3);
fprintf('TEM MODE:\n');
fprintf('TEM mode has no cutoff frequency.\n');
```

```
fprintf('Therefore, TEM can propagate from DC to the operating frequency.\n\n');
```

```
beta_TEM = k0;
```

```
fprintf('TEM Propagation Constant = %.4f rad/m\n\n',beta_TEM);
```

```
m = 1;
```

```
n = 0;
```

```
fc_TE10 = (c/2)*sqrt((m/a)^2+(n/b)^2);
```

```
if f > fc_TE10
```

```
    beta_TE10 = k0*sqrt(1-(fc_TE10/f)^2);
```

```
else
```

```
    beta_TE10 = 0;
```

```
end
```

```
fprintf('TE10 MODE:\n');
```

```
fprintf('Cutoff Frequency = %.3f GHz\n',fc_TE10/1e9);
```

```
if f > fc_TE10
```

```
    fprintf('TE10 mode is PROPAGATING.\n');
```

```
    fprintf('Propagation Constant = %.4f rad/m\n\n',beta_TE10);
```

```
else
```

```
    fprintf('TE10 mode is BELOW CUTOFF.\n\n');
```

```
end
```

```
m = 1;
```

```
n = 1;
```

```
fc_TM11 = (c/2)*sqrt((m/a)^2+(n/b)^2);
```

```
if f > fc_TM11
```

```
    beta_TM11 = k0*sqrt(1-(fc_TM11/f)^2);
```

```
else
```

```
    beta_TM11 = 0;
```

```
end
```

```
fprintf('TM11 MODE:\n');
```

```

fprintf('Cutoff Frequency = %.3f GHz\n',fc_TM11/1e9);
if f > fc_TM11
    fprintf('TM11 mode is PROPAGATING.\n');
    fprintf('Propagation Constant = %.4f rad/m\n\n',beta_TM11);
else
    fprintf('TM11 mode is BELOW CUTOFF.\n\n');
end
x = linspace(0,a,100);
y = linspace(0,b,60);
[X,Y] = meshgrid(x,y);
ETE = sin(pi*X/a);
figure;
surf(X*1000,Y*1000,ETE);
shading interp;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Electric Field');
title('TE10 Mode Electric Field Distribution');
colorbar;
view(2);
ETM = sin(pi*X/a).*sin(pi*Y/b);
figure;
surf(X*1000,Y*1000,ETM);
shading interp;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Electric Field');
title('TM11 Mode Electric Field Distribution');
colorbar;

```

```

view(2);
ETEM = ones(size(X));
figure;
surf(X*1000,Y*1000,ETEM);
shading interp;
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Normalized Electric Field');
title('TEM Mode Field Distribution');
colorbar;
view(2);
modes = {'TEM','TE10','TM11'};
beta = [beta_TEM beta_TE10 beta_TM11];
figure;
bar(beta);
set(gca,'XTickLabel',modes);
ylabel('Propagation Constant (rad/m)');
title('Propagation Constant Comparison');
grid on;
fc = [0 fc_TE10 fc_TM11]/1e9;
figure;
bar(fc);
set(gca,'XTickLabel',modes);
ylabel('Cutoff Frequency (GHz)');
title('Cutoff Frequency Comparison');
grid on;
fprintf('-----\n');
fprintf('SIMULATION COMPLETED SUCCESSFULLY\n');
fprintf('-----\n');

```

OUTPUT



